Netflix Discovery: Scaling Personalization

Applying Matrix Factorization to the 100M+ Rating Prize Dataset

TECHNICAL OVERVIEW & RESULTS

Information Overload & Sparsity

The Data Challenge

Netflix provided a catalog of 17,770 movies and 480,000 users. With over 100 million ratings, the scale is massive, yet the user-item interaction matrix is **>98% sparse**.

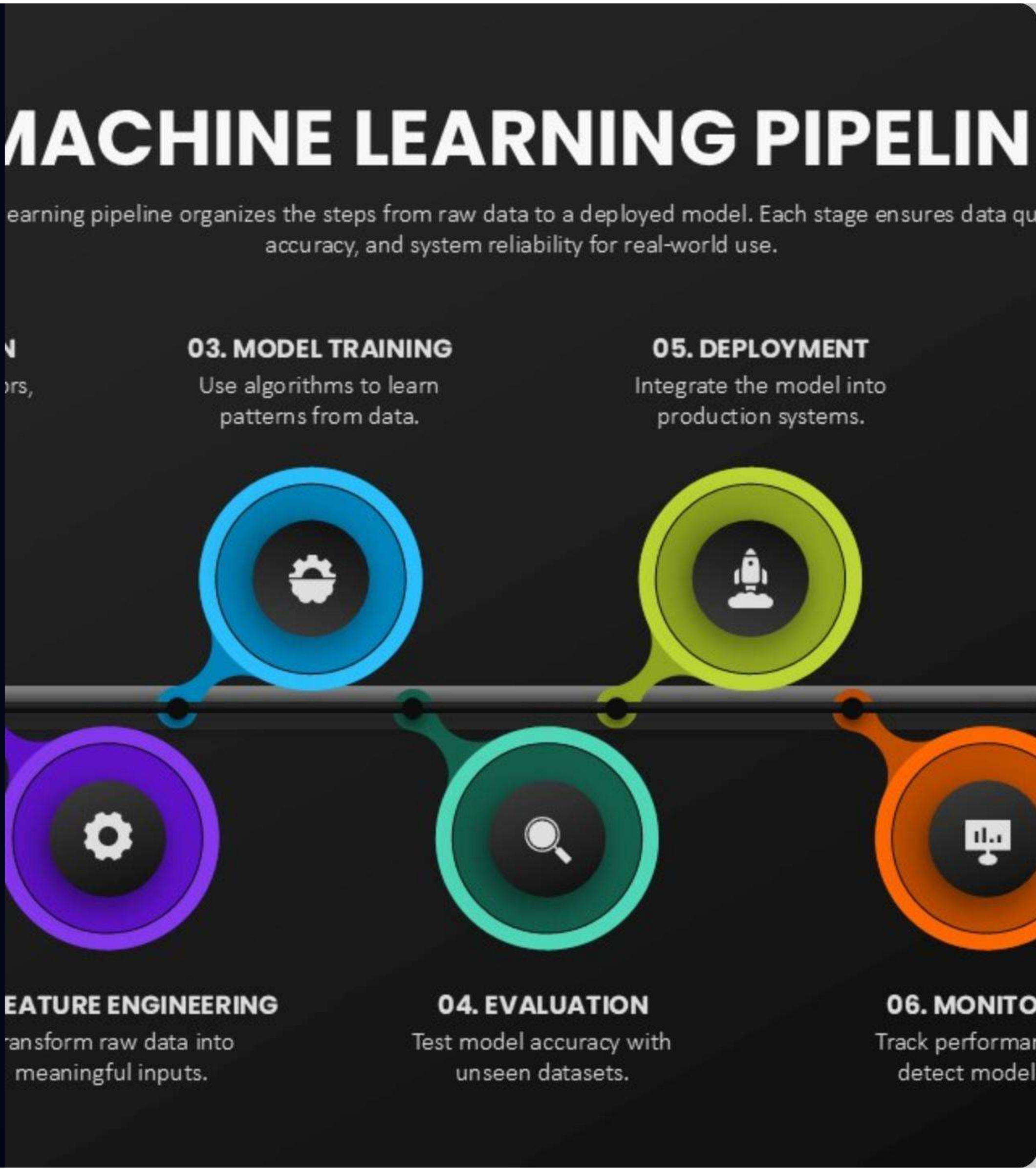
The Business Goal

Improve the "Cinematch" RMSE baseline of 0.9525. A 10% improvement translates to significantly better user retention through highly accurate content discovery.

The Analytical Pipeline

Our strategy focused on transforming raw, noisy text logs into a high-density training environment:

- **High-Density Sampling:** Filtered for users with ≥50 ratings to ensure rich collaborative signal.
- **Temporal Splitting:** Used an 80/20 chronological split to mimic production (predicting future watches).
- **ID Re-mapping:** Normalized user and movie indices for memory-efficient computation.

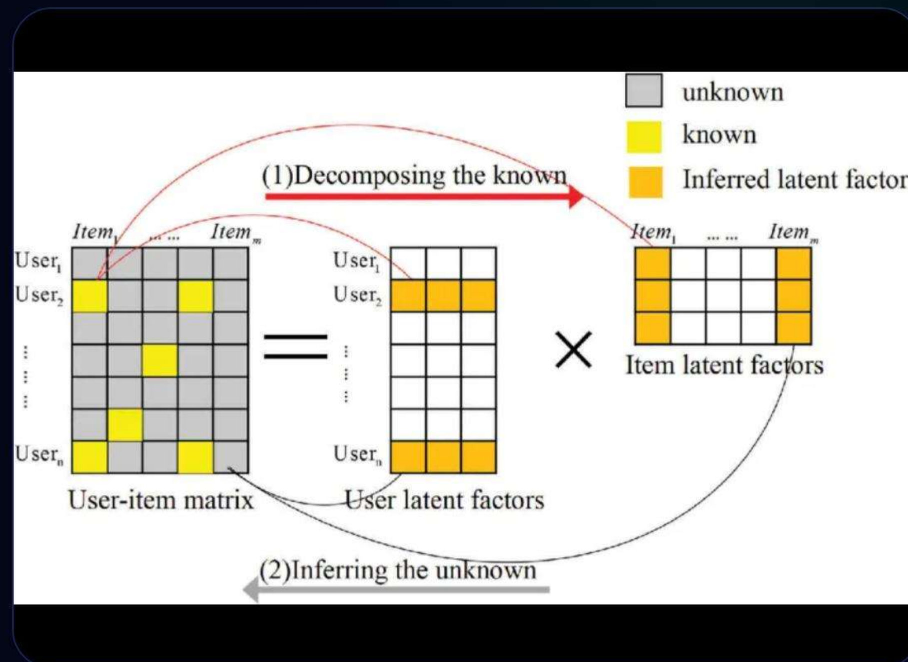


Matrix Factorization vs. UBCF

Latent Factor Discovery

Funk-SVD: Decomposes the interaction matrix into dense 100-dimensional vectors. It explicitly models user and item biases to handle "strict" or "lenient" raters.

User-Based CF: A heuristic approach using Cosine Similarity. While intuitive, it struggles with the high sparsity of the Netflix catalog.



Uncovering Hidden Biases

Positive Rating Skew

EDA revealed a mean rating of ~3.7. Users self-select movies they expect to enjoy, creating a "positive bias" that models must account for to avoid over-prediction.

Power Law Distributions

Both users and movies follow a long-tail distribution. A small group of "super-raters" dominates the signal, while niche content lacks sufficient co-rating density.

Accuracy Benchmarking

SVD demonstrates superior predictive precision, outperforming User-Based methods by over 11% in RMSE. Latent factors successfully bridge the sparsity gap where neighborhood methods fail.

Precision at Scale: MAP@10

- Success (Error ≤ 0.5★): 68%
- Moderate (0.5–2.0★): 25%
- High Error (≥ 2.0★): 7%

Ranking quality (MAP@10) confirms that SVD effectively sorts relevant content (ratings ≥ 3.5) into the Top-10 positions, significantly reducing the "discovery fatigue" for users.

Personalized Discovery Feeds

The Feed

Top-10 recommendations for a sample user, featuring a blend of blockbusters and long-tail niche titles.

Success Case Analysis

User 1488844 Predictions:

- Shawshank Redemption: 4.62 (Actual: 5★)
- Pulp Fiction: 4.38 (Actual: 4★)
- The Godfather: 4.29 (Actual: 4★)

Insight: SVD captures the "Classic Cinema" latent factor for this user with <0.4 absolute error.